

Introduction

How This Course Will Run

About the instructor

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Education

- PhD student in Electrical Engineering — CINVESTAV Unidad Guadalajara (*current*)
- MC in Electrical Engineering — CINVESTAV Unidad Guadalajara
- BS in Mechatronics Engineering — Instituto Tecnológico de Tepic

Research interests

- Computer architecture
- IA models for edge computing
- Hardware/software co-design
- Architectures for AI computing
- VLSI digital design verification

Agenda

- **What is NLP?**
- **The two pillars of this course:** embeddings and transformers
- **How the course runs:** theory paired with notebooks
- **Evaluation:** two lab assignments
- **Logistics:** the course's GitHub repository

What is NLP?

Natural Language Processing, briefly

Natural Language Processing (NLP) is the area of AI concerned with getting computers to work with human language — text and speech — in a way that's useful: understanding it, generating it, or acting on it.

You already use NLP every day:

- **Machine translation** — Google Translate, DeepL
- **Search engines** — matching a query's *meaning*, not just its keywords
- **Virtual assistants & chatbots** — Siri, ChatGPT
- **Sentiment analysis** — is this review positive or negative?
- **Summarization** — condensing a long article into a few sentences

Why it's a hard problem

Language is ambiguous

- *"I saw her duck."* — verb or noun?
- Sarcasm, idioms, tone

Language is high-dimensional

- Huge vocabularies
- Meaning depends on order and context

Computers need numbers

- Text isn't a number — it has to become one
- *How* you turn words into numbers determines how much meaning survives

Key idea. Most of this course is really about *representation*: how to turn words and sentences into numbers a neural network can compute with, without losing their meaning.

The Two Pillars of This Course

Embeddings and Transformers

This course covers two main topics, each building on the last:

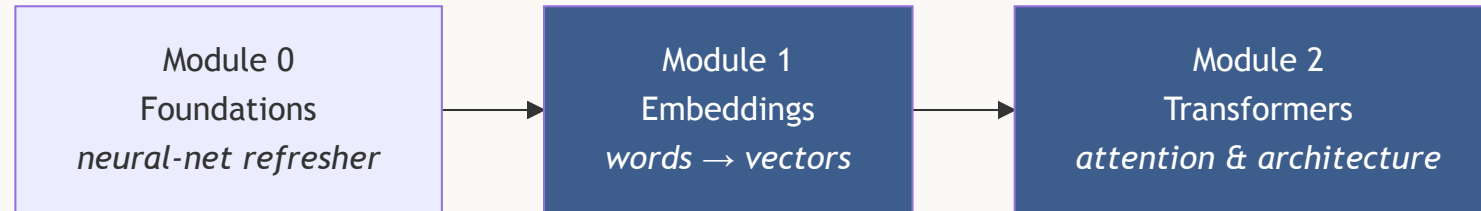
1 · Embeddings

- Turning words (and documents) into vectors
- From heuristic pooling to recurrent encoders

2 · Transformers

- Attention as a better way to combine information
- The Transformer architecture end to end
- The backbone of every modern LLM

The course roadmap



Module 0 refreshes the neural-network basics both pillars assume; Module 1 builds the representations, Module 2 builds the architecture that made them famous.

How the Course Runs

Theory + notebooks, side by side

For **each** of the two main topics, the material comes in two matched parts:

Theory

- A slide deck per topic
- Intuition first plus math
- Builds the vocabulary and the "why"

Notebooks

- Hands-on practice.
- Consolidates the theory by *doing*
- Runs locally or on Google Colab

Key idea. The slides explain *why* something works; the notebooks make you *use* it. Neither replaces the other.

Evaluation

Two lab assignments

Evaluation is based on **two lab assignments** (homework), one tied to each main topic:

Assignment 1 — Embeddings

Assignment 2 — Transformers

- Each assignment is a take-home notebook that extends what was practiced in class
- Submit your completed notebook; it will be **reviewed**.

Logistics

Everything lives on GitHub

All course material is distributed through a **GitHub repository** — slides and notebooks alike.

- Material is uploaded **incrementally**, as each topic is ready
- You'll always find the **latest slides and notebooks** in the repo
- Pull the repository regularly to stay up to date
- Assignments are submitted back through the same workflow

Let's get started

Module 0 refreshes the neural-network foundations.